

Topology-Aware & Hierarchical All-Reduce; Libraries

- Libraries:
 - NCCL (GPU-centric), Gloo (CPU/GPU), MPI (HPC)
- Algorithms:
 - Ring: bandwidth-optimal for large messages
 - Tree: lower latency for smaller messages
 - Reduce-scatter + all-gather: core of sharded training
- Hierarchical all-reduce:
 - Intra-node reduce-scatter (NVLink)
 - Inter-node reduce-scatter (IB/Ethernet)
 - Inter-node all-gather
 - Intra-node all-gather
- Practice: bucket sizes, algorithm selection per tensor size, topology hints

Tensor Parallelism & Sharding (Megatron, ZeRO/FSDP)

Tensor parallel (intra-layer):

- Split attention heads or MLP dimensions across GPUs
- Per-layer collectives: all-gather/all-reduce or reduce-scatter in fwd/bwd
- Works best within fast NVLink/NVSwitch “islands”

Optimizer/gradient/parameter sharding:

- Stage 1: shard optimizer states
- Stage 2: shard optimizer + gradients
- Stage 3: shard optimizer + gradients + parameters (optionally offload to CPU/NVMe)

Reduces per-rank memory; introduces extra comm for gathers/scatters

FSDP (PyTorch):

- Fully sharded per module; parameters/optimizer states are sharded
- All-gather param shards before compute; reduce-scatter grads after
- Configurable wrap/granularity, mixed precision, and offload

Trade-off:

- Big memory relief enables very large models
- More frequent and fine-grained communication—tune bucket sizes, prefetch, and overlap
- Combine with activation checkpointing and BF16/FP16